1. Student Performance Analysis

```
import numpy as np

student_scores = np.array([
    [85, 78, 92, 74],
    [90, 88, 85, 80],
    [76, 95, 89, 72],
    [88, 84, 94, 86]
])

subjects = ["Math", "Science", "English", "History"]

# Calculate average scores
average_scores = np.mean(student_scores, axis=0)

# Display average scores
for i in range(len(subjects)):
    print(subjects[i], "Average:", average_scores[i])

# Find the subject with the highest average
highest_index = np.argmax(average_scores)

print("\nSubject with the highest average:", subjects[highest_index])
print("Highest average score:", average_scores[highest_index])
```

Output:

```
Math Average: 84.75
Science Average: 86.25
English Average: 90.0
History Average: 78.0

Subject with the highest average: English
Highest average score: 90.0
```

2. Average Price of Products

```
import numpy as np


# Sales data for 3 products, with 3 recorded prices for each product
sales_data = np.array([
    [100, 150, 200],
    [250, 300, 350],
    [400, 450, 500]
])


# Calculate the average price of all products
average_price = np.mean(sales_data)


print("Average price of all products:", average_price)
```

Output:

```
Average price of all products: 300.0
```

3. House Data Analysis — Houses with More Than 4 Bedrooms

```
import numpy as np

# House data: [Bedrooms, Square Footage, Sale Price]
house_data = np.array([
    [3, 1500, 250000],
    [5, 2500, 450000],
    [4, 2000, 350000],
    [6, 3000, 550000],
    [5, 2800, 500000]
])

# Select houses with more than four bedrooms
selected_prices = house_data[house_data[:, 0] > 4, 2]

# Calculate average sale price
if len(selected_prices) > 0:
    average_price = np.mean(selected_prices)
    print("Average sale price:", average_price)
else:
    print("No houses with more than four bedrooms found.")
```

Output:

```
Average sale price: 500000.0
```